

Student Result & Grade Management System

Proposal

Selected Scenario: Student Result Processing App

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1. Introduction

The Student Result & Grade Management System is a desktop enterprise application designed to digitalize and automate the academic performance and the tracking process within a university. The system targets key administrative points: manual result entry errors, delayed grade reporting, and the absence of centralized student academic records.

The selected major scenario is Student Result Processing software covering the entire process from student and subject registration, result entry by lecturers, grade calculation, and the decision making of pass and fail and GPA distributions.

The application will be developed using Java (NetBeans IDE), MySQL as the relational database backend, and Jasper Reports for report generation, aligning with the preferred toolset for EAD-1.

2. Requirements

2.1 Input Requirements

- **Student Registration Data**
 - o Obtain student ID, full name, date of birth, pathway, and batch year for creating and maintaining individual student profiles.
- **Subject/Module Configuration**
 - o Input subject codes, subject names, credit hours, and associated semester/year, calculating the grading structure each result is mapped against.
- **Examination Result Entry**
 - o Accept raw marks for each student per subject, including components such as coursework, mid-term, and final examination scores.
- **User Authentication Credentials**
 - o Receive login inputs (username and password) to control role-based access for Admin, Lecturer, and Student roles.

2.2 Process Requirements

- **Grading Mechanism**
 - o Automatically calculate total marks, percentage, letter grade (A/B/C/D/F), and grade points based on customizable grading rules stored in the database.
- **GPA/CGPA Calculation**
 - o Compute semester GPA and cumulative GPA per student using credit weighted grade point averages across all enrolled subjects.
- **Pass/Fail and Repeat Determination**
 - o Evaluate pass/fail status per subject and overall semester status; mark students who require repeat examinations based on institutional threshold rules.

- **Data Validation and Exception Handling**
 - o Validate all mark entries (range 0–100), include a powerful integrity between students and subjects, and handle exceptions such as duplicate entries or missing records without throwing an error.

2.3 Output Requirements

- **Individual Student Result Sheet**
 - o Generate a per-student result report listing all subjects, marks, grades, GPA, and semester standing, formatted for official distribution using Jasper.
- **Batch Performance Summary Report**
 - o Produce a report across multiple students and subjects showing grade distributions, class averages, highest/lowest scores, and pass rate serving as the main university decision making report (joining the Students, Results, and Subjects tables).
- **At-Risk Student Notification Output**
 - o Display and export a filtered list of students who have failed one or more subjects or whose GPA falls below the usual GPA standard.

3. Conclusion

The Student Result & Grade Management System addresses a clearly defined institutional need through a well-scoped scenario: automated, accurate, and role-controlled result processing software. The system showcases OOP techniques to govern the grade computation logic, multithreading can be applied to batch report generation, IO streams manage data export, and exception handling ensures data integrity throughout.

The MySQL normalized data store, combined with Jasper multi-table reports, directly fulfills the Data Store and Reports rubric criteria. The final outcome will include data input, transaction, and output screens along with a functional dashboard, making this a efficient and easily accessible enterprise application.